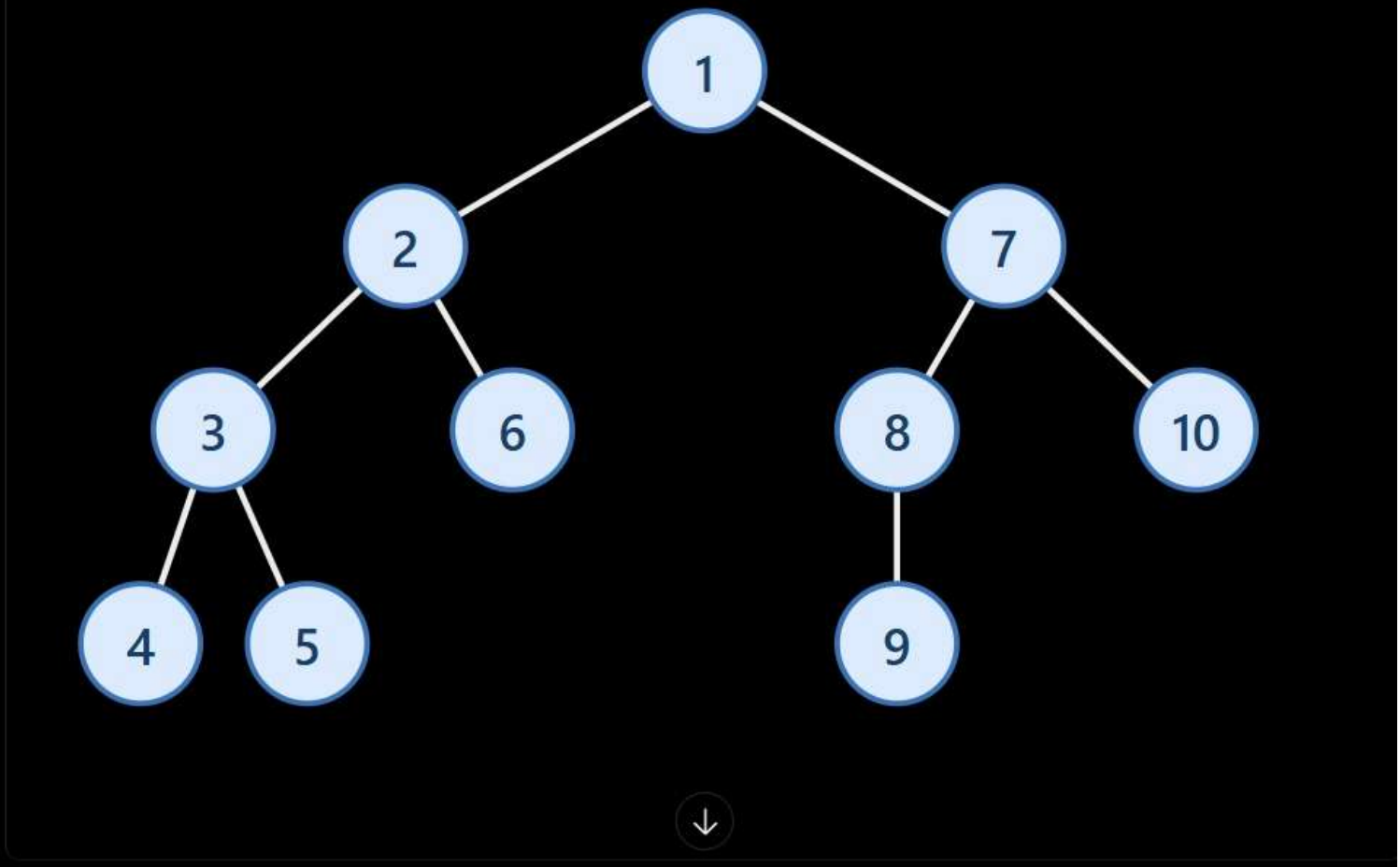
Question 2: BFS and DFS

Diagram



Programiz Python Online Compiler

main.py | Visualize | Run ▶

```python

from collections import deque

graph = {
    1: [2, 7],
    2: [3, 6],
    3: [4, 5],
    4: [],
    5: [],
    6: [],
    7: [8, 10],
    8: [9],
    9: [],
    10: []
}

def bfs(start):
    visited = {start}
    queue = deque([start])
    while queue:
        node = queue.popleft()
        print(node, end=" ")
        for neighbour in graph[node]:
            if neighbour not in visited:
                visited.add(neighbour)
                queue.append(neighbour)
```

Output

```
BFS:
1 2 7 3 6 8 10 4 5 9
DFS:
1 2 3 4 5 6 7 8 9 10
```

```
def dfs(node):
    print(node, end=" ")
    for neighbour in graph[node]:
        dfs(neighbour)

print("BFS:")
bfs(1)

print("\nDFS:")
dfs(1)
print()
```